

Introduction to Nested Queries in DBMS

Name : P KARTHIKEYA

Roll No : 23B81A0521

Section : CSE-A

What are Nested Queries?

- • A nested query (subquery) is a query within another SQL query.
- • The inner query provides data to the outer query.
- • Often used in WHERE, FROM, or SELECT clauses.
- Example:
 - SELECT name
 - FROM students
 - WHERE marks > (
 - SELECT AVG(marks)
 - FROM students
 -);

Types of Nested Queries

- • Single-row subqueries
 - - Returns one row
 - - Operators: =, <, >, etc.
- • Multiple-row subqueries
 - - Returns multiple rows
 - - Operators: IN, ANY, ALL
- • Correlated subqueries
 - - Subquery depends on outer query values

Correlated vs Non-Correlated Subqueries

- Correlated Subquery:
- SELECT name
- FROM students s1
- WHERE marks > (
 - SELECT AVG(marks)
 - FROM students s2
 - WHERE s1.class = s2.class
-);
- Comparison:
 - • Depends on outer query? – Correlated: Yes, Non-Correlated: No
 - • Runs once per outer row? – Correlated: Yes, Non-Correlated: No
 - • Performance – Correlated: Slower, Non-Correlated: Faster

Use Cases of Nested Queries

- Filtering based on aggregates
- Complex joins via subqueries
- Data validation
- Existence checks using EXISTS or NOT EXISTS

Summary & Best Practices

- Nested queries are powerful but can affect performance.
- Prefer non-correlated when possible.
- Always test subquery output independently.
- Use EXISTS for efficiency in some cases.